Activity

1) CCTV image contains salt and pepper noise due to transmission errors.

Technique used : median filter

Reason : median filtering removes black and white noise pixels effectively while preserving the edges. It improves the image quality without blurring important details.

2) A satellite image is corrupted by gaussian noise from sensors.

Technique used : Guassian filter.

Reason : Gaussian filtering smooths the image and reduces gaussian noise caused by sensors. It produces a cleaner image for futher analysis

3). A medical x-ray image has random noise but fine edges must be preserved.

Technique used : Bilateral filter

Reason: Bilateral filtering removes noise while preserving important edges. This helps doctors views bones and tissues more clearly.

4). A scanned document has black and whi[te] dots scattered randomly.

echnique used : median Filter.
Reason : median filter removes salt and pepper noise without affecting the text. It improves the readability of the scanned document.

) A low-light image shows heavy grain without clear boundaries.

Technique used : non-local means (NLM) denoising.
Reason : NLM removes heavy grain noise while preserving image details and edges. It produces a clearer image with better visual quality.

) A foggy road image has very low contrast.

Technique used : Histogram equalization.
Reason : Histogram equalization improves image contras by spreading intensity values. It makes roads and obj more visible in foggy conditions.

) An underwater image appears dull and washed out.
Technique used : contrast stretching
Reason : contrast streching enhances brightness a color contrast. It makes underwater objects appe clearer and more visible.

8) A chest X-Ray shows poor visibility of internal structures.

Technique used: CLAHE (Contrast limited Adaptive histogram Equalization).

Reason: CLAHE improves local contrast without over enhancing noise.

9) A night time traffic image lacks intensity variation.

Technique used: Histogram Equalization.

Reason: Histogram Equalization increases intensity variation and enhances the visibility of vehicles and roads at night.

10) A grayscale image uses only a narrow intensity range.

Technique used: Contrast stretching

Reason: Contrast stretching expands the narrow intensity range to the full range making the image clearer and improving overall contrast.